Kritansh Uppal

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EDUCATION

Bachelor of Technology in Computer Science & Engineering *Aug 2024 – Present*
Punjab Engineering College, Chandigarh **CGPA: 8.82/10**

EXPERIENCE

TBRL, DRDO — Research Intern *May 2026 – Jul 2026*

- Built a **computer vision pipeline** for automated blast-fragment kinematic analysis using **high-speed imagery**, enabling fragment detection, **multi-object tracking**, and **velocity estimation**.
- Developed motion-estimation pipelines using **YOLO**, **ByteTrack**, **optical flow**, and geometric analysis to extract **projectile trajectories** from high-frame-rate recordings.
- Researching **impact-energy estimation** from target-surface damage by integrating **crater segmentation**, **depth reconstruction**, and **physics-based energy modeling** of fragment impacts.

Zimo One — Software Intern *Jun 2025- Jul 2025*

- Resolved **15+ production bugs** in Node.js backend services, improving internal tool uptime by **12%** for workflows used by **30+ team members**.
- Implemented REST API validation using schema-based checks, reducing regression issues by **25%** across multiple services.

PROJECTS

SkillForge AI - Adaptive Onboarding Engine *Mar 2026*
FastAPI, React, NLP, LLMs, NetworkX, SentenceTransformers, PDF Parsing Source Code

- Built an **AI onboarding engine** that converts resumes and job descriptions into **personalized learning roadmaps** using a multi-stage pipeline (parsing, skill extraction, gap analysis, path generation).
- Implemented **hybrid skill extraction (keyword + SentenceTransformers)** and a **weighted gap analysis system** to identify and prioritize critical skill deficiencies.
- Developed a **dependency-aware learning engine using NetworkX** and deployed via **FastAPI + React**, delivering explainable outputs through structured JSON reasoning.

Automatic Short-Form Video Generator (Vizard-AI Replica) *Dec 2025 – Jan 2026*
Computer Vision, Multimodal Systems, Python Source Code

- Architected an automated pipeline to generate short-form clips from long videos, reducing manual editing time by **70%**.
- Compared uniform vs scene-change based frame sampling and selected adaptive key-frame extraction, improving clip relevance by **22%**.
- Processed **5+ hours of video per run** using reproducible CV-based frame extraction.

Intelligent Data Extraction from Business Reports (1st / 50+ Teams, PECathon 2025) *Nov 2025*
Computer Vision, Document AI, YOLO

- Led the architecture of a document intelligence system to extract structured insights from unstructured business PDFs.
- Achieved **90%+ table and chart detection accuracy** using layout segmentation and image preprocessing.
- Awarded **1st place** for superior system reliability, extraction accuracy, and end-to-end automation.

GrowPlay – AI-Powered Table Tennis Coaching Platform *Jan 2026 – Present*
React, FastAPI, MongoDB, Firebase, YOLO, RAG

- Built an AI-powered table tennis coaching platform using React, FastAPI, MongoDB, and Firebase Authentication, featuring role-based architecture with Super Admin support and scalable foundations for future Admin, Coach, and Player portals.
- Developed a sports-specific RAG assistant that combines table tennis literature and player performance data to generate personalized coaching recommendations, match analysis, and training insights.
- Integrated a computer vision pipeline with YOLO-based table detection, automated coordinate extraction, player pose estimation, and ball-tracking analytics for performance evaluation and tactical feedback.

TECHNICAL SKILLS

ML / AI: Machine Learning, Computer Vision, Object Detection, RAG Pipelines, LLMs, Feature Engineering, NLP
Frameworks & Libraries: Scikit-learn, LangChain, SentenceTransformers, OpenCV, FAISS, NumPy, Pandas, Matplotlib
Backend & Databases: FastAPI, Firebase, MongoDB, REST APIs, SQL
Frontend: React, Streamlit
Programming: Python, C++, SQL, JavaScript
Tools & Systems: Git, Linux, YOLO, ByteTrack, RTMPose

ACHIEVEMENTS

- **Winner, PECathon 2025**- Open-to-all inter-college hackathon conducted during PECFEST with 50+ teams.
- **Awarded by Punjab Governor Sh. Gulab Chand Kataria ji** for excellence in sports.
- **State Gold Medalist**- Table Tennis, Chandigarh.

LEADERSHIP EXPERIENCE

- **Creative Head, WOPI**- Led branding and visual strategy for an international academic conference.
- **Sub-Head, PECFEST**- Coordinated logistics for 10+ events and managed 30+ volunteers serving 900+ attendees at the college's flagship festival.